

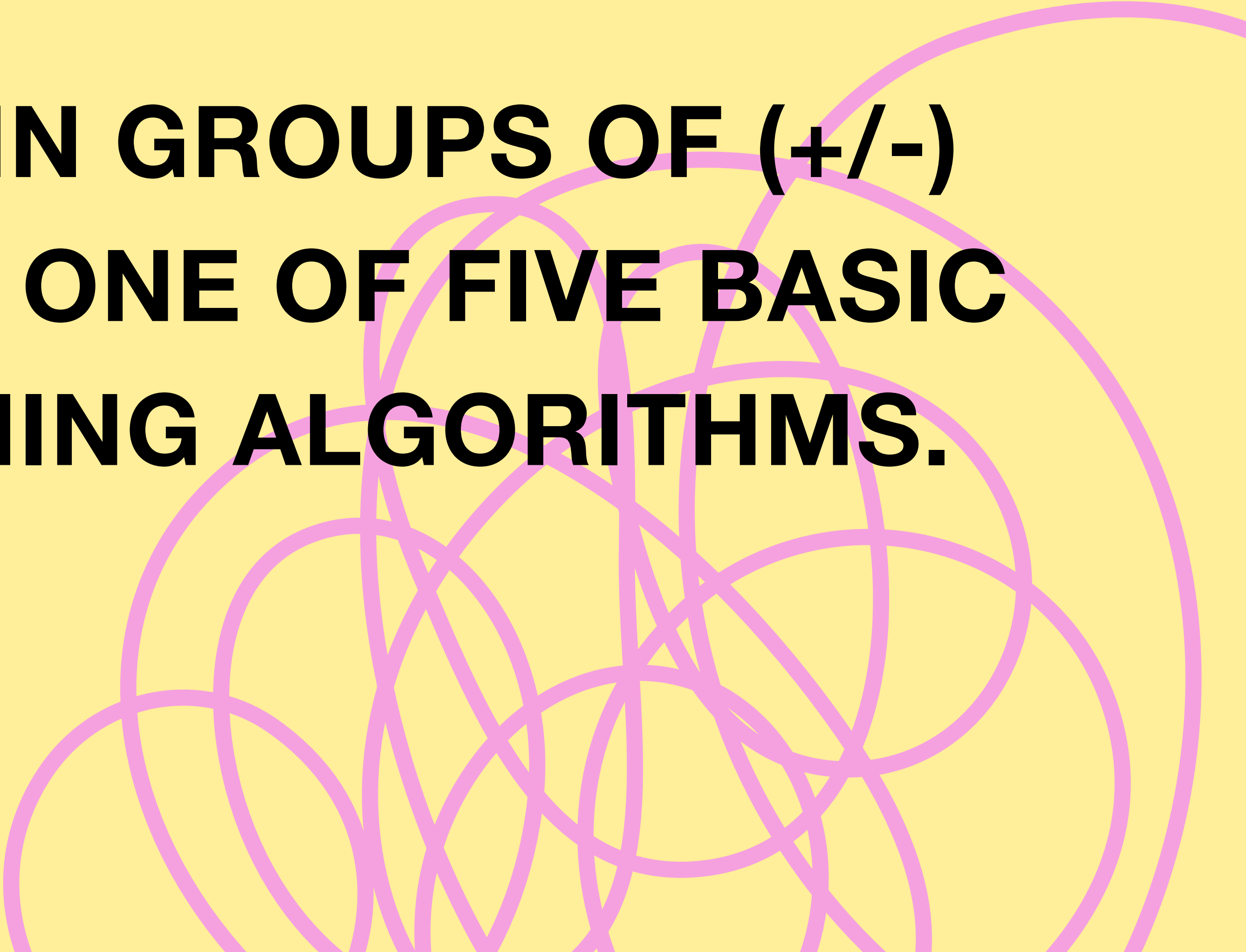


Complex Social &
Computational Systems

BASIC MACHINE LEARNING ALGORITHMS

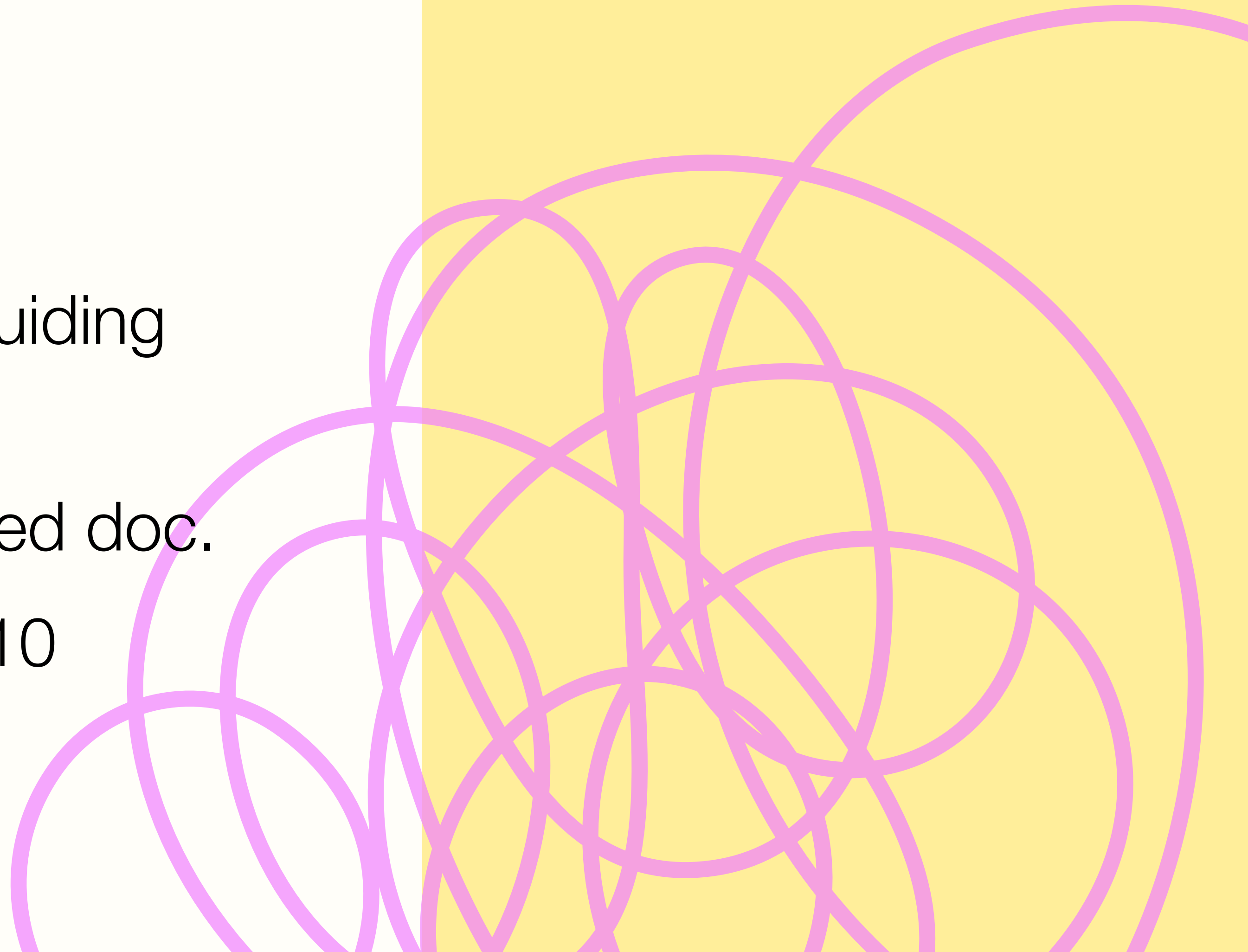
A PSYCHOLOGIST'S GUIDE TO MACHINE LEARNING
ZURICH, MARCH 16-20, 2026

**GET TOGETHER IN GROUPS OF (+/-)
THREE TO STUDY ONE OF FIVE BASIC
MACHINE LEARNING ALGORITHMS.**

A decorative graphic consisting of several overlapping, hand-drawn style purple circles of varying sizes, located in the bottom right corner of the slide.

FROM 09:30 TO 11:00

- ▶ Search for information online.
- ▶ I provide links as starting points.
- ▶ There are specific and general guiding questions for each algorithm.
- ▶ Collect your results in the provided doc.
- ▶ Prepare a short presentation of 10 minutes for your colleagues.





**WHAT ALGORITHMS
DO WE HAVE?**

LOGISTIC REGRESSION

CLASSIFICATION

What types of logistic regression are there?

In what way is logistic regression related to linear regression?

What differences are there?

What is the sigmoid function? What is it also called?

What is the Maximum Likelihood Estimation method?



SUPPORT VECTOR MACHINES CLASSIFICATION

What is a support vector?

What is a hyperplane?

What does linearly separable vs. non-linearly separable data mean?

What is a kernel?

What is a regularization parameter?



DECISION TREES

CLASSIFICATION

What are nodes, edges, and leaves?

What are the Gini coefficient, entropy, and information gain?

What do CART and ID3 stand for?

What are early stopping and pruning?

Advanced decision tree algorithms: What are a random forest, bagging, and boosting?



K-MEANS CLUSTERING

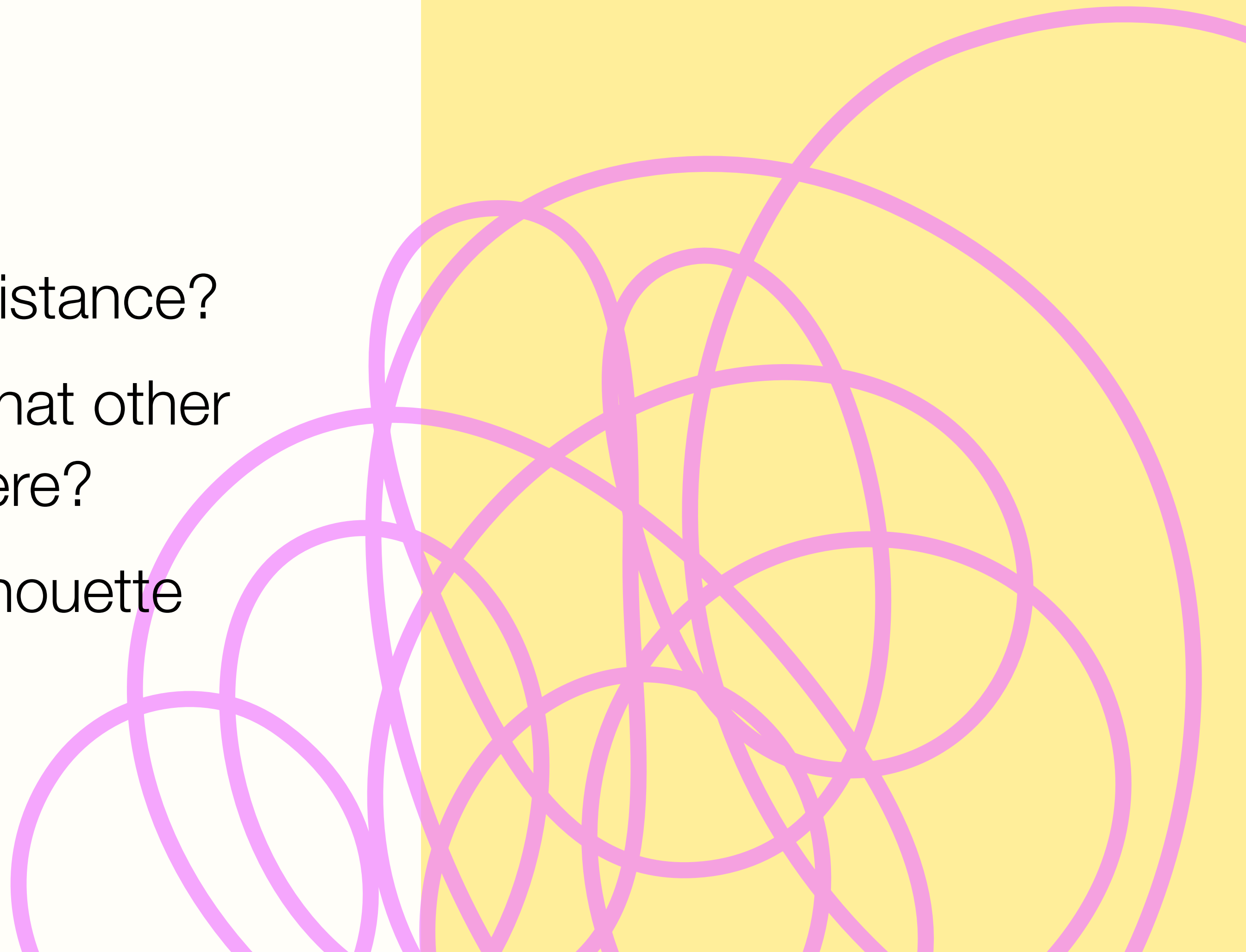
What does k-means stand for?

What is a centroid?

What is inter-cluster vs. intra-cluster distance?

What is the Euclidean distance and what other distance or similarity measures are there?

What is the elbow method and the silhouette coefficient?



PRINCIPAL COMPONENT ANALYSIS

DIMENSIONALITY REDUCTION

What is meant by the curse of dimensionality?

What is a covariance matrix?

What is an eigenvector, a principal component?

What is an eigenvalue and how does it relate to the variance in the data?

What is a scree plot?



A FEW FINAL NOTES

- ▶ You will find all questions, information and links in the respective document online. The documents are linked in the GitHub repository of this course.
- ▶ The general questions always apply to assess the suitability of a machine learning algorithm.
- ▶ The specific questions include key words that you can look out for during your research.
- ▶ You do not have to answer all questions.
- ▶ You might want to look for videos that contain helpful visualizations to understand the algorithms.

TAKE YOUR OWN BREAK.
PRESENTATIONS AT 11:00.